CHEMISTRY REVISION SHEET (CLASS X)

Sindh Textbook Board Karachi — Key Concepts, Equations & Formulae

1. Chemical Equilibrium & Stoichiometry

- **Number of Moles:** $\text{Moles} = \text{Given Mass (g)} / \text{Molar Mass (g/mol)}$.
- **Avogadro's Constant:** $1\ \text{Mole} = 6.022 \times 10^{23}\ \text{particles}$.
- **Equilibrium Constant (K_c):** For reaction $aA + bB \rightleftharpoons cC + dD$, $K_c = [C]^c[D]^d / ([A]^a[B]^b)$.

2. Acids, Bases & Salts

Concept	Formula / Scale	Indicator / Note
pH Calculation	$pH = -\log_{10}[H^+]$	Acidic when $pH < 7$
pOH Calculation	$pOH = -\log_{10}[OH^-]$	Basic when $pH > 7$
Water Constant	$pH + pOH = 14$	At 25°C standard condition

3. Organic Chemistry Functional Groups

Class	General Formula	Example
Alkanes (Saturated)	C_nH_{2n+2}	Methane (CH_4), Ethane (C_2H_6)
Alkenes (Double Bond)	C_nH_{2n}	Ethene (C_2H_4)
Alkynes (Triple Bond)	C_nH_{2n-2}	Ethyne (C_2H_2)
Alcohols	$R-OH$	Methanol (CH_3OH)

4. Environmental Chemistry & Solutions

- **Molarity (M):** $M = (\text{Moles of solute}) / (\text{Volume of solution in Liters})$.
- **Hardness of Water:** Caused by dissolved bicarbonates, sulfates, and chlorides of Calcium and Magnesium.